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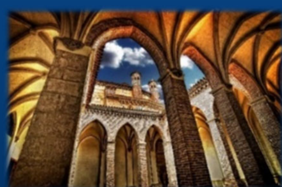

SPICE models for electrical simulation of commercial MOSFET arrays ALD1105/06/07

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***Abstract*— In this work, several parameter extraction procedures were performed on commercial Mosfets ALD1105 and ALD1106/07. The data were fit to find a suitable Spice model for their use on electrical simulations. Since Mosfet Spice level 1 parameters have a direct physical meaning, the presented methodology represents an introductory tool for Mosfet modeling. Moreover, these models can be used for circuit development at transistor level. Simulation results are in good agreement with experimental data.**

Keywords— CMOS, Spice models, circuit design.

I. Introduction

Nowadays, electronics circuit development is strongly related to CAD tools; exhaustive simulations and circuit optimization techniques must be done carefully in order to get a successful design. In this sense, the use of reliable device models is very important, since it increases the possibility of successful implementations reducing time and cost. Recently, the use of commercial MOSFET arrays for circuit development has become an excellent option due to the high cost of manufacturing a full custom CMOS integrated circuit [1]. Additionally, its good matching properties allow the development of accurate current mirrors and other basic analog blocks [2-4]. However, in the case of commercial MOSFET arrays ALD110X, the manufacturer does not provide an accurate device model [5]. Thus, several extraction techniques for MOSFET basic physical parameters were performed, all data were fitted and associated to Spice model Level 1 as a tool for accurate and reliable simulations.

II. Threshold Voltage and Transconductance Parameter Extraction

First, an electrical characterization must be achieved over the MOSFET devices. The transconductance curve in saturation was measured using Keysight B2902A sources. The circuit configuration depicted in Fig. 1 was considered in order to obtain the plots. In this case, V_{GS} was swept from 0V to 4V with 0.1V steps and V_{DS} was fixed to 5V to ensure saturation regime, V_{BS}=0. Fig. 2 shows the measurement plot for ALD1105 and ALD1106 NMOS transistors.

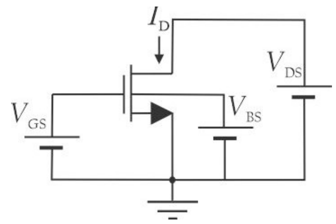

Fig. 1. Circuit configuration with I-V sources.

Considering the drain current in saturation:

$$I_D = \frac{\beta}{2}(V_{GS} - V_{TH})^2 \qquad (1)$$

This equation can be linearized as follows:

$$\sqrt{I_D} = \sqrt{\frac{\beta}{2}}(V_{GS} - V_{TH}) \qquad (2)$$

Where β=k_nW/L is the transistor transconductance; k_n is the transconductance parameter; W the width and L the length of the MOS gate. From the measured data, the threshold voltage V_{TH} can be calculated as the intercept to the x's axis from the V_{GS} vs. sqrt(I_D) plot, [6], Fig. 3. As can be noticed, for NMOS ALD1105 and ALD1106 the threshold voltage is close to 0.6V. Similarly, to compute the transconductance parameter β using (2), the term $\sqrt{(\beta/2)}$ can be calculated as the slope from a linear fit from V_{GS} vs. sqrt(I_D) plot. For this case, the data fulfilling V_{GS}>0.6V are considered for a proper slope computation. Thus, for slope m_1, the transconductance parameter can be calculated using β=2m_1^2, in this case for NMOS transistors the methodology brings β=627µA/V2.

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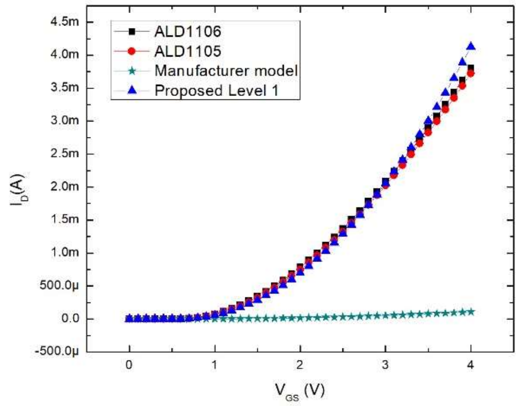

Fig. 2. Comparison of between measurements of ALD1105 and ALD1106 in saturation and simulations of manufacturer Spice model and the proposed Level 1.

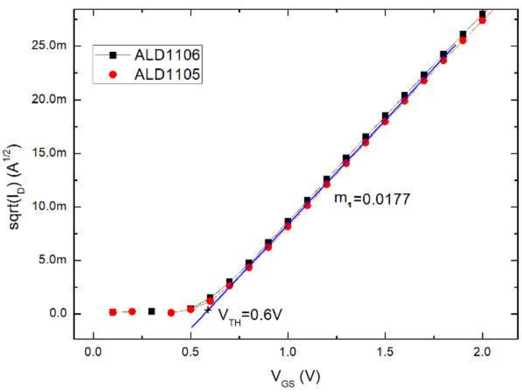

Fig. 3. Extraction of V_{TH} from the transconductance curve in saturation.

III. CHANNEL LENGTH MODULATION PARAMETER

For this parameter, the output characteristic must be measured, thus, a primary sweep of V_{DS} from 0V to 4V with 0.1V steps and a secondary sweep of 1V to 4V with 1V steps were considered and V_{BS}=0. The plot for ALD1105 and ALD1106 is depicted in Fig. 4. The output resistance r_o of a MOS transistor can be calculated using the following expression:

$$\frac{1}{r_o} = \lambda I_D \tag{3}$$

Since (3) is expressed in linear form, the slope of a plot of I_D vs. $1/r_o$ will be the value of λ. For this purpose, $1/r_o$ can be calculated from Fig. 4, however, it is necessary to consider only those data in the saturation region. For this reason, the data in the range 3V⩽ V_{DS} ⩽4V is selected and since the data of both transistors are close enough, the plot of Fig. 5 is considered. From this plot, the slope of a linear fit brings $m_2=\Delta I_D/\Delta V_{DS}=1/r_o$ and the value for I_D is selected in the middle of the range on each trace. Hence, from these four traces, the new plot I_D vs. $1/r_o$ is depicted in Fig. 6, and as stated above, the slope m_2 from a linear fit brings the value of λ in this case a value of 0.0185V^{-1}

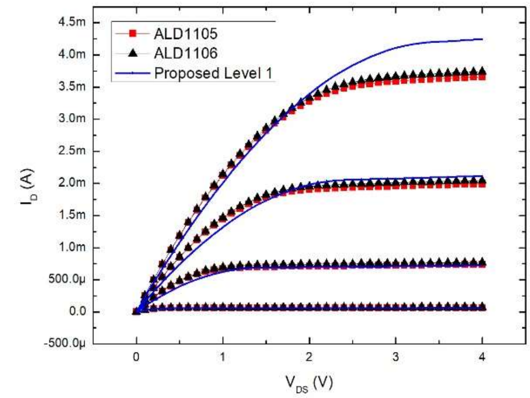

Fig. 4. Measured and simulated NMOS output characteristic.

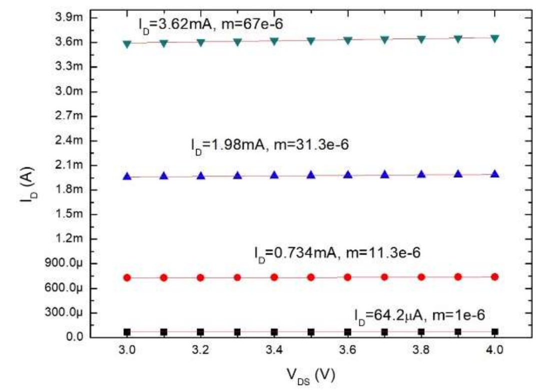

Fig. 5. Linear fit for I_D and $1/r_o$ computation.

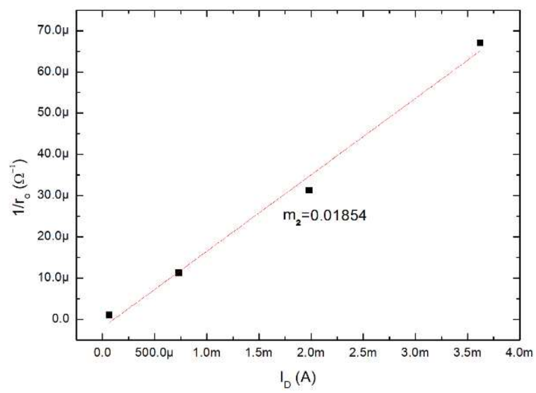

Fig. 6. Linear fit to data, slope m_2=λ.

IV. Bulk Effect Parameter

For the bulk effect parameter γ, the methodology presented in [6] was also followed. In this case, the transconductance curve must be measured in the linear region. A primary sweep of V_{GS} from 0 to 4V with 0.1V steps is considered, a secondary sweep of V_{BS} from -1V to -4V with -1V steps and V_{DS} is fixed to 100mV. The measurement is depicted in Fig. 7, in this case only the measurement results of NMOS ALD1106 are presented, NMOS ALD1105 does not present any observable change in threshold voltage. The dependency of voltage threshold with bulk-source voltage V_{BS} is given by:

$$V_{TH} = V_{TH0} + \gamma\left(\sqrt{(|2\phi_F| - V_{BS})} - \sqrt{|2\phi_F|}\right) \quad (4)$$

Where V_{TH0} is the zero-bias threshold voltage when V_{BS}=0. The term $2\phi_F$ is the surface potential, due to the lack of information related to carrier concentrations N_A or N_D, a default value for $2\phi_F$ in Spice model Level 1 can be used. From (4) as can be noticed, γ can be calculated as the slope m_3 of a plot $\sqrt{(|2\phi_F| - V_{BS})} - \sqrt{(|2\phi_F|)}$ vs. V_{TH}, therefore from Fig. 8, the threshold voltage for each trace at different V_{BS} can be extracted as the intercept to the x's axis and the value of $\sqrt{(|2\phi_F| - V_{BS})} - \sqrt{(|2\phi_F|)}$ must be computed. This data generates the plot shown in Fig. 9. Using the linear fit, the slope m_3 is the value for γ in this case $0.749V^{1/2}$. A comparison of the measurement data and simulation is presented in Fig. 10, the γ extracted fits accordingly to the measurements.

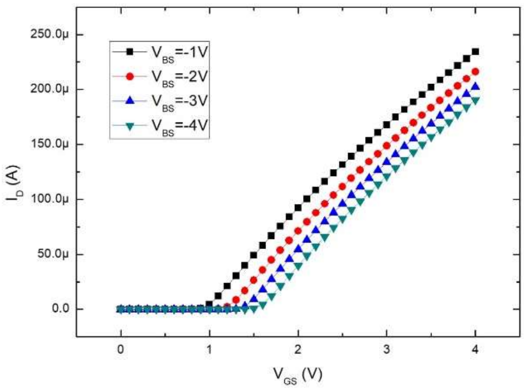

Fig. 7. Transconductance curves in linear region with different V_{BS} voltages.

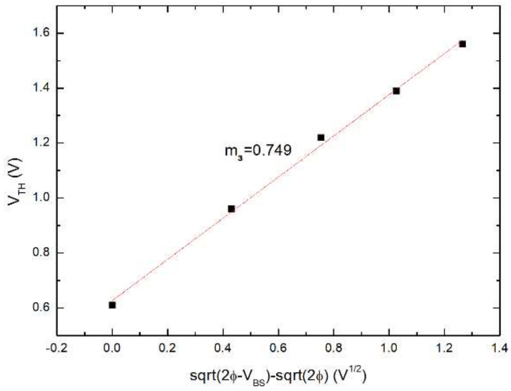

Fig. 8. Linear fit of data, m_3=γ.

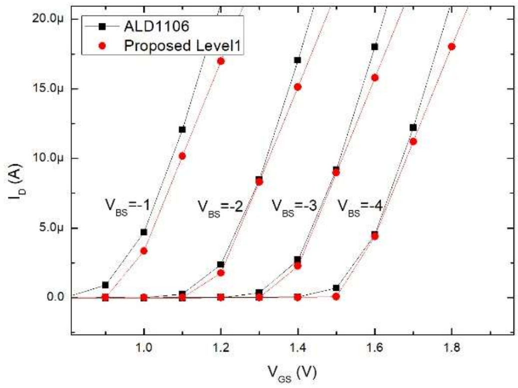

Fig. 9. Comparison between measurements and simulations including γ.

V. Measurement of Gate-Source Capacitance

The MOS transistor parasitic capacitances are not easy to extract, commonly for overlap capacitances, special structures are made on a given technology for proper characterization. In these conditions, we focused on C_{gs} since is the most important capacitance when the transistor is working in the saturation region. A typical C-V characterization curve was not feasible due to the presence of Electrostatic Discharge protective diodes (ESD) at the gate; the size of such diodes represents a large component of the input capacitance. Therefore, in order to measure such input capacitance in saturation, the measurement set depicted in Fig. 10 was used. Input voltage v_i was $100mV_{pp}$ with a voltage offset of 0.8V and V_{DS} was fixed to 3.3V.

Since resistor R is 100kΩ the cutoff frequency at node "x" is related with $C_{gs} + C_{probe}$, the frequency response difference with and without the transistor ALD1106 is depicted in Fig. 11. Using the expression $\omega = 1/\{R(C_{gs}+C_{probe})\}$, such frequency difference is related to a value C_{gs}=3pF.

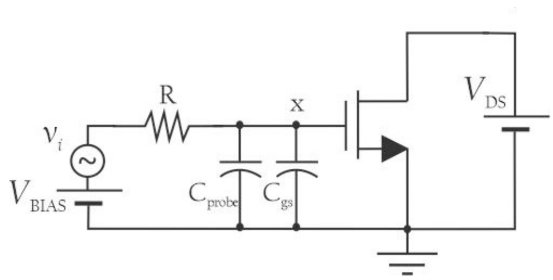

Fig. 10. Configuration used to measure Cgs.

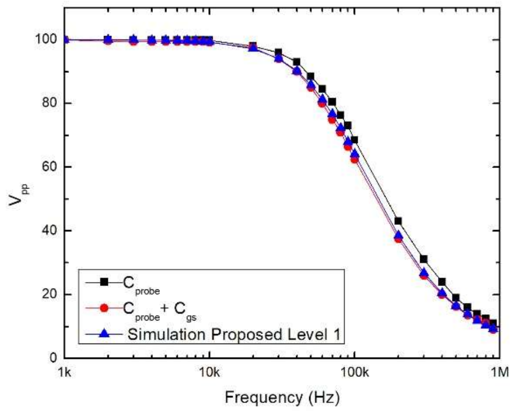

Fig. 11. Measured frequency response at node "x".

VI. MODEL PARAMETERS FOR SPICE LEVEL 1

The extracted parameters must be included in the Spice model Level 1. For the case of KP is necessary to recall from section II the value β=k_nW/L=627μA/V^2. With the purpose of considering a given value for the aspect ratio, sizes of W=50μm and L=10μm were considered, this fact brings the following value for KP=126u.

For the case of C_{gs}, in model 1 as stated in [7], $C_{gs} \approx (2/3)WLC_{ox}$, where C_{ox} is the device oxide capacitance per unit area. Spice computes C_{ox} following $C_{ox} = \varepsilon_0 \varepsilon_{ox}/t_{ox}$, using the relative oxide permittivity and oxide thickness t_{ox}. Therefore, this expression can be used to calculate t_{ox}, hence, the model parameter TOX=4.5E-9 must be included for a proper C_{ox} model computation, other overlap capacitances were set to a minimal default value. Some parameter values were little adjusted manually for optimal performance, the model 1 for the NMOS is as follows:

MODEL ALD1106 NMOS (LEVEL=1
+CBD=0.01p CBS=0.01p CGDO=0.01p
+CGSO=0.01p GAMMA=.7 KP=136u
+TOX=0.9E-9 LAMBDA=0.02 PHI=.9 VTO=0.6
+L=10E-6 W=50E-6)

Simulation results of the model are depicted in Fig. 2, Fig. 4, Fig. 9 and Fig. 11. For PMOS transistors ALD1105 and ALD1107 the same methodology was applied. Simulations of the proposed PMOS model are depicted in Fig. 12 and Fig 13. The PMOS model is:

MODEL ALD1107 PMOS (LEVEL=1
+CBD=0.01p CBS=0.01p CGDO=0.01p
+CGSO=0.01p +GAMMA=.45 TOX=0.9E-9
+KP=100u LAMBDA=0.03 PHI=.8 VTO=-0.6
+L=10E-6 W=20E-6)

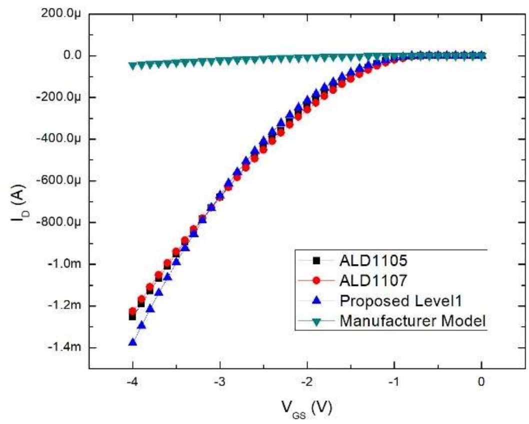

Fig. 12. Measurements in saturation regime and simulations of manufacturer Spice model and the proposed Level 1.

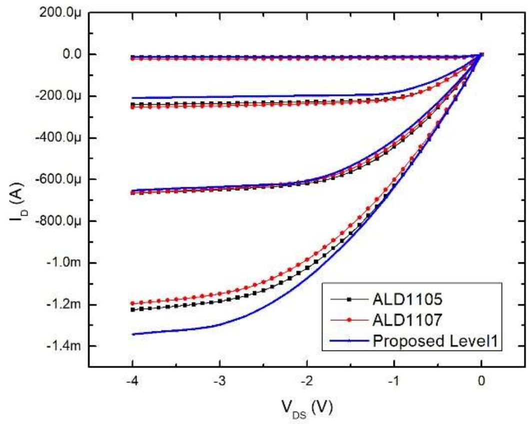

Fig. 13. Measured output characteristics and simulated using proposed PMOS Level 1.

VII. CONCLUSIONS

In this work, the basic extraction procedures in commercial MOSFET arrays ALD1105/06/07 were followed to obtain a Spice model suitable for electrical simulations. Although these commercial devices can work with a supply voltage of 12V, the methodology only considered an operating range of up to 4V since we focused on applications in saturation regime with 3.3V supply voltage. The simulated curves agree fairly with measurements, a statistical characterization, in order to improve such models, will be reported soon.

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